

LISTA FYR, Norway

WMO: 014270

Lat: 58.1090N

Lon: 6.5678E

Elev: 14

StdP: 101.16

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-7.5	-5.5	-13.5	1.2	-5.2	-11.0	1.5	-3.7	18.5	5.5	16.6	5.1	3.9	50	0.778

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.6	22.4	17.1	20.6	16.5	19.3	16.0	18.2	21.0	17.2	19.6	16.5	18.6	6.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.0	12.2	19.3	16.2	11.5	18.5	15.5	11.0	17.8	51.3	21.2	48.6	19.7	46.3	18.6	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.5	14.6	13.1	-8.6	24.6	2.6	1.9	-10.5	26.0	-12.0	27.1	-13.5	28.2	-15.4	29.6		
(5)			-9.6	19.4	2.5	1.1	-11.4	20.2	-12.9	20.8	-14.3	21.4	-16.1	22.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	2.1	1.7	3.1	6.2	9.7	12.6	15.3	15.7	13.4	9.4	5.8	3.2
	DBStd	5.72	3.45	3.45	2.65	2.52	2.86	2.42	2.21	2.11	2.10	2.64	3.16	3.85
	HDD10.0	1234	245	233	214	117	39	3	0	0	2	42	128	211
	HDD18.3	3704	503	467	473	363	269	174	99	86	149	277	376	469
	CDD10.0	583	0	0	0	4	29	81	163	178	103	23	2	0
	CDD18.3	11	0	0	0	0	1	1	4	5	1	0	0	0
	CDH23.3	24	0	0	0	0	1	2	16	7	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
(14) Wind	WSAvg	6.5	7.4	6.4	6.3	5.7	5.9	6.1	5.9	6.0	6.6	7.1	7.0	7.2
(15) Precipitation	PrecAvg	1664	192	140	120	94	76	85	100	139	155	191	182	183
	PrecMax	2137	380	301	264	150	174	194	253	237	297	371	357	341
	PrecMin	1028	11	19	23	43	14	7	14	25	51	52	83	23
	PrecStd	286	82	75	57	31	41	45	52	47	61	82	80	86
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.7	7.6	10.9	16.2	21.5	22.9	25.7	24.2	20.6	15.6	12.4	10.0
		MCWB	7.6	6.5	6.5	10.2	14.3	15.6	18.4	17.8	17.1	13.1	11.6	9.4
	2%	DB	7.8	6.9	7.9	13.5	18.5	20.0	22.4	22.3	18.5	14.4	11.6	9.3
		MCWB	6.8	6.1	5.4	9.1	13.2	15.3	17.3	17.6	15.5	12.6	10.6	8.6
	5%	DB	7.2	6.5	6.8	11.6	16.1	18.2	20.4	20.5	17.3	13.6	10.7	8.7
		MCWB	6.2	5.7	5.4	7.9	12.1	14.6	16.8	16.8	14.8	12.0	9.6	7.8
	10%	DB	6.6	6.0	6.1	10.0	14.1	16.4	18.9	19.2	16.4	13.0	10.0	8.1
		MCWB	5.7	5.3	5.0	7.4	11.0	13.7	16.2	16.2	14.3	11.6	8.8	7.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	7.1	7.3	11.4	16.0	17.2	19.6	19.3	17.8	14.0	12.0	9.6
		MCDB	8.6	7.3	9.0	14.2	19.9	20.0	23.6	22.3	19.4	14.9	12.2	9.8
	2%	WB	7.1	6.5	6.4	9.7	13.9	16.1	18.3	18.2	16.4	13.2	10.7	8.8
		MCDB	7.5	6.8	7.4	12.3	17.4	18.9	21.4	21.0	17.8	13.9	11.3	9.1
	5%	WB	6.5	6.0	5.8	8.6	12.5	15.1	17.3	17.4	15.4	12.5	9.8	8.1
		MCDB	7.0	6.3	6.5	10.9	15.5	17.4	19.7	19.8	16.9	13.3	10.5	8.5
	10%	WB	5.9	5.4	5.2	7.7	11.3	14.0	16.4	16.6	14.7	11.8	9.0	7.3
		MCDB	6.5	5.9	6.0	9.4	13.6	16.1	18.5	18.7	16.1	12.8	9.8	8.0
(35) Mean Daily Temperature Range	5% DB	MDBR	3.4	3.6	3.7	4.6	4.6	4.3	4.3	4.6	4.2	3.9	3.6	3.7
		MCDBR	3.3	3.1	4.2	7.3	8.0	7.5	7.4	6.6	5.6	4.2	3.6	3.0
		MCWBR	3.5	2.8	3.3	4.5	4.9	4.8	4.5	4.1	3.8	3.5	3.9	3.4
	5% WB	MCDBR	3.2	2.6	3.5	6.1	7.2	6.3	6.2	5.6	4.8	3.5	3.5	2.9
		MCWBR	3.5	2.7	2.9	4.1	4.8	4.5	4.2	4.0	3.9	3.4	3.9	3.4
(40) Clear-Sky Solar Irradiance	taub		0.315	0.337	0.354	0.387	0.374	0.365	0.379	0.384	0.363	0.345	0.319	0.293
	taud		2.190	2.276	2.326	2.294	2.404	2.455	2.435	2.418	2.463	2.397	2.264	2.158
	Ebn at Noon		505	685	789	819	860	873	850	817	778	674	502	407
	Edh at Noon		51	74	92	111	106	102	102	97	80	65	47	38
(44) All-Sky Solar Radiation	RadAvg		0.34	0.95	2.29	3.84	5.17	5.80	5.23	3.99	2.59	1.24	0.47	0.23
	RadStd		0.07	0.19	0.24	0.43	0.62	0.53	0.57	0.33	0.27	0.19	0.10	0.03

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-122	N/A	N/A

Nomenclature: See separate page